UNDEF

FADENA
Facultad de Defensa Nacional

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Carrera Lic. En Ciberdefensa

Análisis Matemático I

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Año: 2025

Carga horaria: 64

0. ACOTADO:

SI $\lim_{x \to a} f(x) = 0$, Y g ES UNA FUNCIÓN ACOTADA EN $(a-\varepsilon, a) \cup (a, a+\varepsilon)$, ENTONCES



$$\lim_{x \to a} \underbrace{f(x)}_{\to 0} \cdot \underbrace{g(x)}_{\text{ACOTADA}} = 0$$

EJ: $\lim_{x \to 0} \underbrace{x^2}_{\to 0} \cdot \underbrace{\operatorname{sen}\left(\frac{1}{x}\right)}_{\text{ACOTADO}} = 0$

EJ: $\lim_{x \to 0} \underbrace{\operatorname{sen}(x)}_{\to 0} \cdot \underbrace{\frac{1}{\cos\left(\frac{1}{x}\right)+3}}_{\text{ACOTADO}} = 0$

$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1 \Rightarrow 3-1 \leq 3+\cos\left(\frac{1}{x}\right) \leq 4$

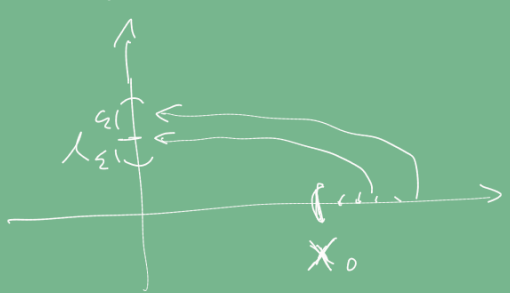
$2 \leq 3+\cos\left(\frac{1}{x}\right) \leq 4$

$\frac{1}{4} \leq \frac{1}{3+\cos\left(\frac{1}{x}\right)} \leq \frac{1}{2}$

LÍMITES EN EL INFINITO:

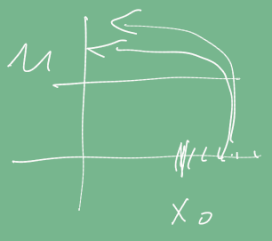
DEF: $\lim_{x \to +\infty} f(x) = \ell \in \mathbb{R}$ SI

$(\forall \varepsilon > 0)\ (\exists x_0 \in \mathbb{R}) / (\forall x \geq x_0)\ |f(x) - \ell| < \varepsilon$



DEF. $\lim_{x\to+\infty} f(x) = +\infty$ SI

$(\forall M>0)\ (\exists x_0 \in \mathbb{R}) / (\forall x \geq x_0) \quad f(x) > M$



DEF. $\lim_{x\to+\infty} f(x) = -\infty$ SI

$(\forall M<0)\ (\exists x_0 \in \mathbb{R}) / (\forall x \geq x_0) \quad f(x) < M$

DEF. $\lim_{x\to-\infty} f(x) = \ell \in \mathbb{R}$ SI

$(\forall \varepsilon>0)\ (\exists x_0 \in \mathbb{R}) / (\forall x \leq x_0) \quad |f(x) - \ell| < \varepsilon$

DEF. $\lim_{x\to-\infty} f(x) = +\infty$ SI

$(\forall M>0)\ (\exists x_0 \in \mathbb{R}) / (\forall x \leq x_0) \quad f(x) > M$

DEF. $\lim_{x\to-\infty} f(x) = -\infty$ SI

$(\forall M<0)\ (\exists x_0 \in \mathbb{R}) / (\forall x \leq x_0) \quad f(x) < M$

Ej: $\lim_{x\to+\infty} \frac{x^3+3x+1}{2x^4+2x^2+1} = \lim_{x\to+\infty} \frac{x^3\left(1+\frac{3}{x^2}+\frac{1}{x^3}\right)}{x^4\left(2+\frac{2}{x^2}+\frac{1}{x^4}\right)} =$

(IND. $\frac{\infty}{\infty}$)

$= \lim_{x\to+\infty} \frac{1}{x} \cdot \frac{1+\frac{3}{x^2}+\frac{1}{x^3}}{2+\frac{2}{x^2}+\frac{1}{x^4}} = 0 \cdot \frac{1}{2} = \boxed{0}$

($\frac{1}{x} \to 0$, $+\infty$; numerador $\to 1$; denominador $\to 2$; cociente $\to \frac{1}{2}$)

Ej: $\lim_{x\to+\infty} \frac{\sqrt{9x^2+6}}{5x-1} = \lim_{x\to+\infty} \frac{\sqrt{x^2\left(9+\frac{6}{x^2}\right)}}{x\left(5-\frac{1}{x}\right)} =$

$= \lim_{x\to+\infty} \frac{\sqrt{x^2}\cdot\sqrt{9+\frac{6}{x^2}}}{x\cdot\left(5-\frac{1}{x}\right)} = \lim_{x\to+\infty} \frac{|x|\cdot\sqrt{9+\frac{1}{x^2}}}{x\cdot\left(5-\frac{1}{x}\right)} =$

$= \lim_{x\to+\infty} \frac{\frac{1}{x}\cdot\sqrt{9+\frac{1}{x^2}}}{\cancel{x}\cdot\left(5-\frac{1}{x}\right)} = \frac{\sqrt{9}}{5} = \boxed{\frac{3}{5}}$

($\frac{1}{x}\to 0$, $\frac{1}{x^2}\to 0$, $\cancel{x}\to 1$)

Ej: $\lim_{x\to-\infty} \frac{\sqrt{9x^2+6}}{5x-1} = \lim_{x\to-\infty} \frac{|x|\cdot\sqrt{9+\frac{1}{x^2}}}{x\cdot\left(5-\frac{1}{x}\right)} = \lim_{x\to-\infty} \frac{\cancel{-x}\cdot\sqrt{9+\frac{1}{x^2}}}{\cancel{x}\cdot\left(5-\frac{1}{x}\right)} =$

$= \lim_{x\to-\infty} -\frac{\sqrt{9+\frac{1}{x^2}}}{5-\frac{1}{x}} = -\frac{\sqrt{9}}{5} = -\frac{3}{5}$

($\frac{1}{x^2}\to 0$, $\frac{1}{x}\to 0$)

Ej: $\lim_{x\to+\infty} x^5 - x^6 + \sqrt{x} = \lim_{x\to+\infty} x^6\cdot\left(\frac{1}{x} - 1 + \frac{\sqrt{x}}{x^6}\right) =$

$= \lim_{x\to+\infty} x^6\cdot\left(\frac{1}{x} - 1 + \frac{1}{x^{\frac{11}{2}}}\right) = -\infty$

($\frac{1}{x}\to 0$, $\frac{1}{x^{11/2}}\to 0$, $\left(\frac{1}{x} - 1 + \frac{1}{x^{11/2}}\right)\to -1$, $x^6\to+\infty$)

Ej: $\lim_{x\to-\infty} x - \sqrt{x^2+1} = -\infty$

($x\to-\infty$, $\sqrt{x^2+1}\to+\infty$)

Ej: $\lim_{x\to+\infty} x - \sqrt{x^2+1} = \lim_{x\to+\infty} \frac{\left(x-\sqrt{x^2+1}\right)\cdot\left(x+\sqrt{x^2+1}\right)}{x+\sqrt{x^2+1}} =$

($x\to+\infty$, $\sqrt{x^2+1}\to+\infty$)

(INDET. $\infty-\infty$)

$= \lim_{x\to+\infty} \frac{x^2-\left(\sqrt{x^2+1}\right)^2}{x+\sqrt{x^2+1}} = \lim_{x\to+\infty} \frac{x^2-(x^2+1)}{x+\sqrt{x^2+1}} =$

$= \lim_{x\to+\infty} \frac{x^2-x^2-1}{x+\sqrt{x^2+1}} = \lim_{x\to+\infty} \frac{-1}{x+\sqrt{x^2+1}} = \boxed{0}$

($x+\sqrt{x^2+1}\to+\infty$)

Ej: $\lim_{x\to+\infty} \frac{5^x+2}{3^x+9} = \lim_{x\to+\infty} \frac{5^x\left(1+\frac{2}{5^x}\right)}{3^x\left(1+\frac{9}{3^x}\right)} =$

$= \lim_{x\to+\infty} \left(\frac{5}{3}\right)^x \cdot \frac{1+\frac{2}{5^x}}{1+\frac{9}{3^x}} = +\infty$

$\left(\frac{5}{3}\right)^x \to +\infty$; $1+\frac{2}{5^x} \to 1$; $1+\frac{9}{3^x} \to 1$

Ej: $\lim_{x\to-\infty} \frac{5^x+2}{3^x+9} = \boxed{\frac{2}{9}}$

$5^x \to 0$; $3^x \to 0$

Ej: $\lim_{x\to+\infty} \frac{\ln(x)}{x} = \lim_{x\to+\infty} \frac{1}{x} \cdot \underbrace{\ln(x)}_{\text{ACOTADO}} = \boxed{0}$

$\frac{1}{x} \to 0$

Ej: $\lim_{x\to+\infty} \frac{5x^3-2x^2+1}{3x^3+x-11} = \lim_{x\to+\infty} \frac{x^3\left(5-\frac{2}{x}+\frac{1}{x^3}\right)}{x^3\left(3+\frac{1}{x^2}-\frac{11}{x^3}\right)} = \frac{5}{3}$

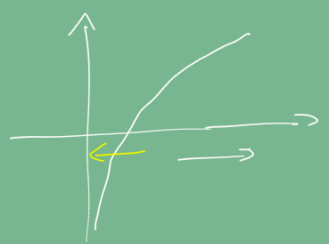
$5-\frac{2}{x}+\frac{1}{x^3} \to 5$; $3+\frac{1}{x^2}-\frac{11}{x^3} \to 3$

Ej: $\lim_{x\to-\infty} \frac{-5x^3+2x^2}{3x^2+x} =$

$= \lim_{x\to-\infty} \frac{x^3}{x^2} \cdot \frac{-5+\frac{2}{x}}{3+\frac{1}{x}} = \lim_{x\to-\infty} x \cdot \frac{-5+\frac{2}{x}}{3+\frac{1}{x}} = +\infty$

$x \to -\infty$; $\frac{-5+\frac{2}{x}}{3+\frac{1}{x}} \to -\frac{5}{3}$

Ej: $\lim_{x\to+\infty} \ln(x) = +\infty$



Ej: $\lim_{x\to+\infty} \ln\left(\frac{1}{x}\right) = -\infty$

$\frac{1}{x} \to 0^+$

$\lim_{x\to+\infty} -\underbrace{\ln(x)}_{\to+\infty} = -\infty$